



32 Xbox 360



36 Memoirs of Memory



41 Antelligence!

Digital Passion

Fuelling the pursuit of technology knowledge

Lead Feature

Studio 64

Enter the
psychedelic
world of the
all-64-bit PC

Nimish Chandiramani

You probably felt special when you bought that new AMD64. Thirty-two extra bits to play with! The sun shone brighter, the sky cleared up, and the heavens smiled upon you as you transcended the dregs of the 32-bit x86 world to new heights of über-geekdom. Your 64-bit desktop would be the Ferrari to your neighbour's bullock cart.

Or would it?

For all the hype that surrounded the launch in 2003 of affordable 64-bit processors for the desktop, all it turned out to be was little more than a big tease. Developments in software have notoriously lagged behind those in hardware, so while we rubbed our hands in glee at the spanking-new hardware platform, we were still using 32-bit software that hasn't been designed to take advantage of 64-bit computing—and we still are.

What's The Bit Idea?

If you've been scratching your head all this while with all this bit-talk, cease worrying immediately. The 32-bit and

64-bit in processor-talk signify the number of bits that the processor uses when it talks to the system memory. Each byte in memory has a unique *address*, denoted by a number that is 32 bits long for a 32-bit processor, and 64 bits long for a 64-bit processor. With two possibilities for each of the 32 bits (0 or 1), 32-bit processors have the ability to access 2^{32} memory locations, each a byte in size—totalling up to just 4 GB of memory. This means it can load 4 GB of code and/or data into your system's RAM (assuming you have that much) and take advantage of the much higher speeds that it offers when compared to hard disks. Now before you say "But 4 GB ought to be enough for anyone!", take a

Illustration Chaitanya Surpur



Illustration Pravin Warhokar